Funding the Fall

Crypto Perpetual Futures Carry + Tail-Risk Strategy

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Summary

A delta-neutral strategy that harvests cross-venue perpetual futures funding rate spreads and shorts coins vulnerable to liquidation cascades. Funding was positive, but square-root price impact, taker fees, and liquidation losses consumed the portfolio.

- **Funding Rate Carry (85% base allocation):** delta-neutral cross-venue spread harvesting across 7 exchanges
- **Liquidation Cascade Overlay (up to 15%):** Merton jump-diffusion + OI/depth risk scoring for tail-risk shorts
- **Universe:** BTC, ETH, SOL, HYPE, DOGE across Hyperliquid, Lighter, OKX, Kraken, Binance, Bybit, dYdX

Sharpe Ratio	Max Drawdown	Total Return
–0.82	**–97.0%**	**–96.4%**

+$39K funding collected | $1.0M in costs (impact + fees + liq.) | **The carry signal works; execution must improve**

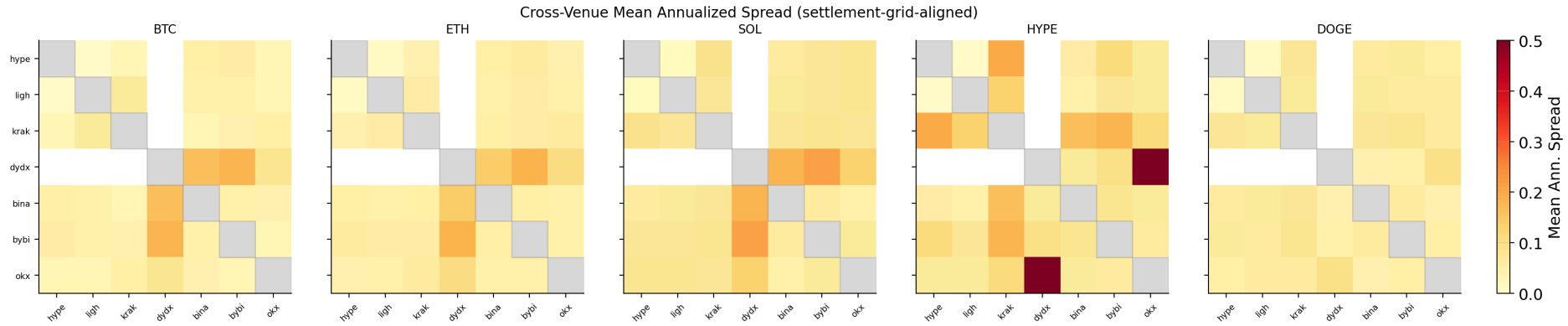
The Opportunity

Funding Rate Dispersion

- Perpetual futures use **funding rates** to anchor prices to spot
- Rates vary 5 to 60%+ annualized across venues
- **Structural cause:** venue fragmentation, different user bases, margin rules
- Top mean spreads: HYPE 61%, SOL 21%, BTC 18%

Cascade Fragility

- High leverage creates **liquidation feedback loops**
- Price drop → forced selling → further drop
- OI/depth ratios reveal structural vulnerability
- HYPE 1048x, BTC 329x, ETH 81x, SOL 69x



How It Works

Carry Leg (85% baseline)

1. Scan 7 venues for funding rate spreads every 8h
2. Enter delta-neutral pairs above optimized threshold
3. Collect net funding payments (long low-rate, short high-rate)
4. Exit on mean-reversion or max holding period
5. 4x leverage per leg

Cascade Leg (up to 15%)

1. Calibrate Merton jump-diffusion on hourly returns
2. Compute OI/depth risk score per coin
3. Short the most fragile coins when score is elevated
4. 1.5x leverage, 24h rebalance

Entry: spread ≥ 35% ann. **Exit:** spread ≤ 1% ann. **Risk score:** $1 - e^{-\mathrm{OI}/(\mathrm{depth} \cdot 200)}$

Risk Controls

Allocation shifts dynamically with aggregate cascade risk: carry scales down up to 30%, cascade scales up, cash buffer grows.

5x

Gross Leverage Cap

Total notional across all positions may not exceed 5x NAV. Hard

40%

Single-Exchange Cap

No more than 40% of NAV exposed to any single exchange.

10%

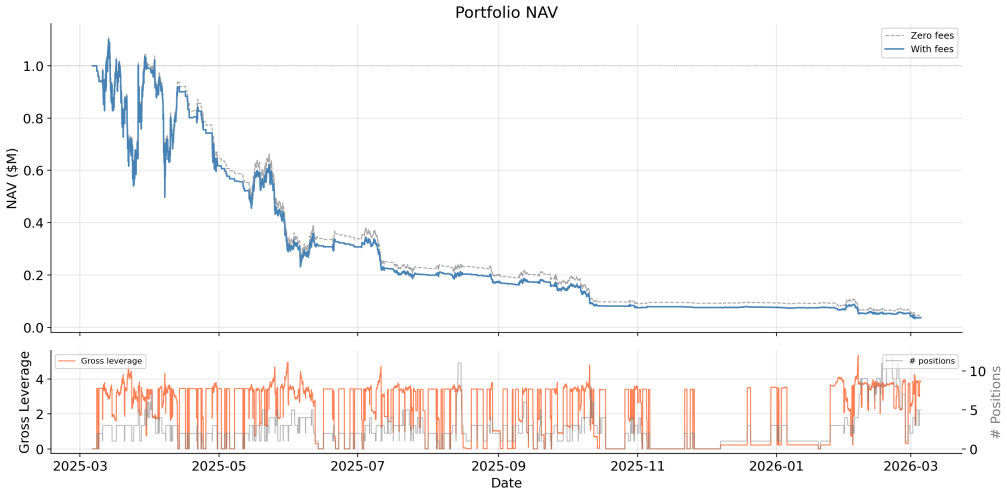
Net Delta Limit

Net directional exposure capped at 10% of NAV. Ensures the

Risk Regime	Carry	Cascade	Cash
Low (score < 0.3)	85%	0%	15%
Medium (0.3 to 0.7)	75%	10%	15%
High (score > 0.7)	60%	15%	25%

Backtest Results

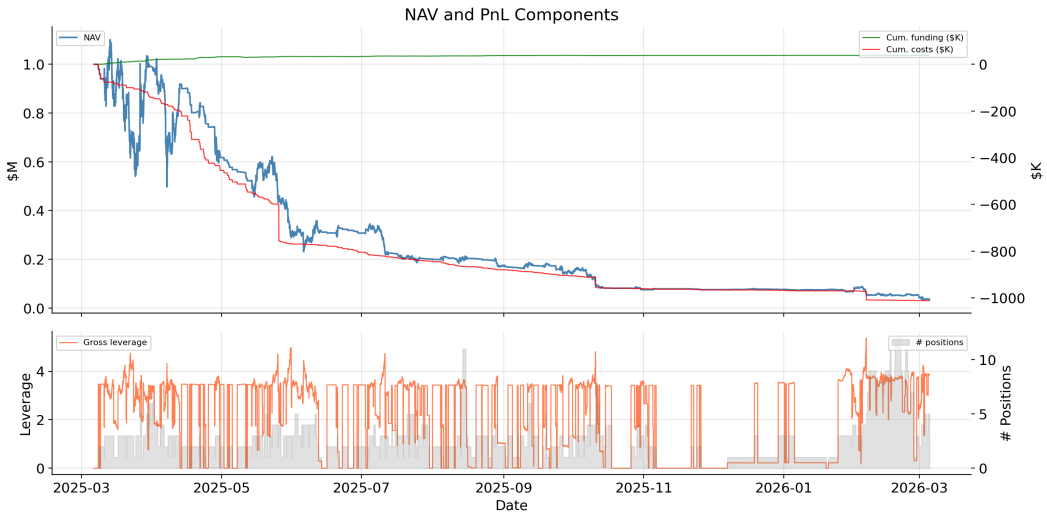
Metric	Impact Only*	Full Costs
Total Return	– 95.4%	– 96.4%
Ann. Volatility	123.3%	124.2%
Sharpe Ratio	– 0.81	– 0.82
Max Drawdown	– 96.3%	– 97.0%



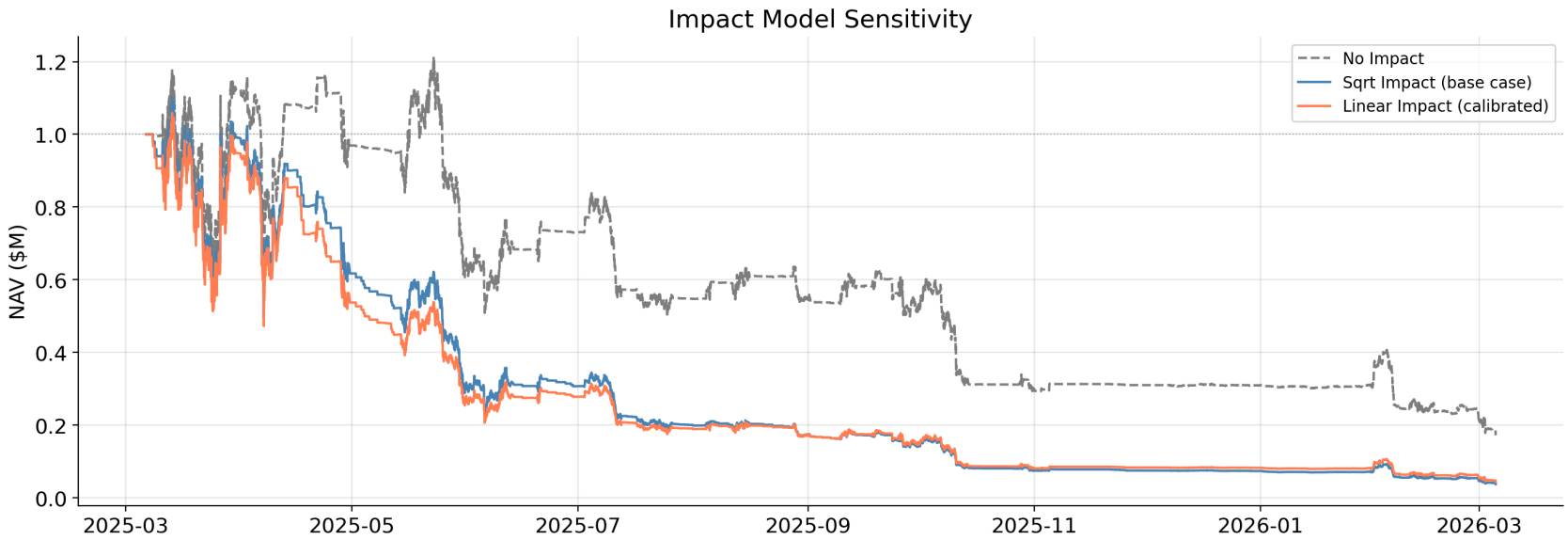
12 months (Mar 2025 to Mar 2026) · 8,737 epochs · $1M initial · 627 trades
*Impact Only = sqrt price impact deducted from cash (no taker fees)

Where the Money Went

Component	PnL
Funding Collected	+$39K
Price PnL	+$10K
Price Impact	– $680K
Taker Fees	– $96K
Liquidation Losses	– $236K
Net PnL	**– $964K**



- 621 carry trades, $200M notional turnover
- Price impact was 67% of total costs ($680K)
- Impact ⟶ smaller positions ⟶ liquidations ($236K)
- Costs were 26x funding income



Sqrt (base case) vs. linear (calibrated per-coin to match sqrt at median trade size) vs. no impact. Impact deducted from cash in real time.

What We Learned

What Worked

- Funding income was positive (+$39K): the carry signal correctly identified real cross-venue spreads
- Pre-cost Sharpes of 0.4 to 2.5 across coins
- Cascade scoring flagged the right coins (HYPE, BTC)
- Risk limits held throughout: no leverage breach, no concentration breach

What Failed

- Price impact dominated: $680K (67% of costs) from moving illiquid books
- Impact → NAV bleed → margin calls: $236K in liquidation losses
- 4x leverage on thin venues generated $200M notional turnover
- Cascade leg underutilized: only 3 trades in 12 months

The thesis is valid. A production version needs maker-only routing, longer holding periods, and lower leverage.

Key Risks

Risk	Mitigant
Funding rate compression	Diversification across 21+ venue pairs reduces single-spread dependence
Exchange counterparty risk	40% single-exchange cap; 7-venue diversification; isolated margin per position
Cascade model mis-specification	Conservative sizing (max 15% NAV); model validated on historical cascade events
Liquidity stress	Square-root impact model; positions sized to available depth; cash buffer in high-risk regimes

Conclusion

The Signal

- Funding spreads are structural, not transient: top pairs averaged 18 to 61% annualized
- Pre-cost Sharpes of 0.4 to 2.5 across five coins
- OI/depth ratios (16x to 1048x) correctly identify cascade fragility
- Delta-neutral carry collected +$39K over 12 months

The Fix

- **Thicker venues:** trade only where depth supports the position size, reducing impact
- **Longer holding periods:** let funding accumulate before exiting
- **Lower leverage:** 2x halves notional turnover per dollar of margin
- **Maker-only routing:** 0bps on Hyperliquid and Lighter eliminates taker fees

Funding rate carry in crypto perps is a real inefficiency.

This backtest demonstrates signal validity and quantifies the execution threshold for profitability.